El Sewedy Electrometer · Education Initiative · 2026

ElSewedy GenZ Coders

Phase 1 Curriculum Guide

5 Courses · Session-by-Session Breakdown · Internal Reference · 2026

8	1.5 hrs	4	4 hrs
Core Sessions per Course	Per Core Session	Support Sessions per Course	Per Support Session

#	Course	Age	Price	Deliverable
1	**Scratch**	Age 10+	**500 EGP**	*A fully designed original game*
2	**Intro to C++**	Age 13+	**600 EGP**	*A working quiz / mini app*
3	**Advanced C++**	Age 14+	**700 EGP**	*A complete playable game (SFML)*
4	**Electronics & Robot**	Age 13+	**600 EGP**	*A built and coded robot*
5	**Build Web App with AI**	Age 13+	**500 EGP**	*A live deployed web application*

"Build skills. Track progress. Launch careers." — ElSewedy GenZ Coders

Scratch

Ages 10+ · 500 EGP · Visual Programming · Project: Original Game

8 Core Sessions
1.5 hrs each
4 Support Sessions
4 hrs each

Course Overview

Scratch is the entry point for students aged 10 and above. Using a drag-and-drop visual language, students learn core programming concepts — events, loops, conditions, and variables — by building games and animations they are proud to show off.

By the End, Students Will:

- Use events, loops, conditions and variables confidently
- Build a complete, playable game from scratch
- Debug their own projects independently
- Present and explain their work to others

#	Session / Topic	Activity / Project	Goal / Principle
1	Interface + Motion	Move a sprite around the stage	Understand the Scratch environment
2	Events + Loops	Start game & create continuous movement	Control flow and repetition
TS 1	**Explore & Customise**	Free practice — add sprites, backgrounds, sounds	*Build confidence with the tools*
3	Conditions + Variables	Score system + touch detection logic	Decision making and data tracking
4	Catch Game	Player catches falling objects — full mini game	Apply all logic learned so far
TS 2	**Debug & Extend**	Fix score bugs, add sound effects	*Independence and problem solving*
5	Timer + Difficulty	Speed increases as game progresses	Enhance gameplay with challenge
6	Maze Game	Navigate to a goal through obstacles	Combine all blocks into one project
TS 3	**Polish Catch or Maze Game**	Add levels, sounds, winning screen	*Creativity and refinement*
7	Clones + Advanced Logic	Spawn multiple enemies using clones	Advanced logic and performance
8	Final Project	Student designs and builds their own game	Full application of all skills
TS 4	**Final Project Showcase**	Present game to group + give/receive feedback	*Communication and pride in work*

■ Core Session — 1.5 hrs (Engineer-led) ■ Technical Support Session — 4 hrs (CTA-led)

□ Intro to C++

Ages 13+ · 600 EGP · Problem Solving · Project: Quiz / Mini App

8 Core Sessions
1.5 hrs each
4 Support Sessions
4 hrs each

Course Overview

Intro to C++ moves students from visual to text-based programming. They learn the fundamentals of C++ — variables, data types, conditions, loops, and functions — through a sequence of progressively more complex projects, culminating in a working quiz game they designed.

By the End, Students Will:

- Write, compile and run C++ programs confidently
- Use variables, conditions, loops and functions correctly
- Organise code using functions and clean structure
- Debug and explain their own code independently

#	Session / Topic	Activity / Project	Goal / Principle
1	Variables + I/O	Print messages, take user input	Understand program basics
2	Data Types + Calculations	Build a simple calculator	Work with different data types
TS 1	**Variables & Types Lab**	Exercises: fix broken code, extend calculator	*Consolidate fundamentals*
3	If / Else + Logic	Decision-making program — grade checker	Conditions and branching
4	Guess the Number Game	Number guessing with conditions and feedback	Apply logic in a fun context
TS 2	**Extend the Guess Game**	Add hints, attempt limits, replay option	*Deepen understanding through building*
5	Loops	Repeat game attempts using while/for loops	Repetition and control flow
6	Functions	Refactor game into named functions	Code organisation and reuse
TS 3	**Loops & Functions Challenges**	Mini exercises: FizzBuzz, countdown, menu	*Fluency with core concepts*
7	Quiz Game	Multiple-question quiz with scoring	Structure and flow control
8	Final Project + Debugging	Complete mini app, find and fix all errors	Full understanding + independence
TS 4	**Code Review & Demo**	Walk through code with CTA, explain decisions	*Communication and confidence*

■ Core Session — 1.5 hrs (Engineer-led) ■ Technical Support Session — 4 hrs (CTA-led)

□ Advanced C++

Ages 14+ · 700 EGP · OOP + Game Dev · Project: Complete Playable Game

8 Core Sessions
1.5 hrs each
4 Support Sessions
4 hrs each

Course Overview

Advanced C++ takes students deep into Object-Oriented Programming and real-time application development using SFML. By building a fully playable game — with rendering, movement, enemies, collision, score, and health — students graduate with a project that demonstrates genuine engineering skill.

By the End, Students Will:

- Design and implement class hierarchies using OOP principles
- Apply inheritance and polymorphism in a real project
- Build a real-time game loop with SFML
- Deliver a complete, playable game as a final product

#	Session / Topic	Activity / Project	Goal / Principle
1	Classes + Constructors	Design a Player class with attributes and actions	Encapsulation — structure data
2	File Separation + Vectors	Split into .h/.cpp, store multiple objects	Clean code and data management
TS 1	**OOP Exercises**	Build a small standalone class from scratch	*Solidify class design skills*
3	Inheritance	Create Entity base → Player and Enemy classes	Shared behaviour and hierarchy
4	Polymorphism	Virtual update() method across class types	Flexible and extensible design
TS 2	**Inheritance Challenges**	Extend class tree, add new derived types	*Hierarchy and override mastery*
5	SFML — Window + Draw	Set up game loop, render shapes on screen	Real-time application foundation
6	Movement + Enemies	Keyboard input + vector of enemy objects	Apply OOP + data structures together
TS 3	**SFML Lab**	Fix rendering bugs, smooth movement	*Real-time programming confidence*
7	Collision + Game System	Detect player–enemy collision, track score/health	Full gameplay interaction loop
8	Final Game Project	Build and polish a complete playable game	Integration — real product delivery
TS 4	**Game Demo Day**	Play each other's games, give structured feedback	*Pride in work + presentation skills*

■ Core Session — 1.5 hrs (Engineer-led) ■ Technical Support Session — 4 hrs (CTA-led)

Electronics & Build Your Robot

Ages 13+ · 600 EGP · Hardware + Arduino · Project: Working Robot

8 Core Sessions
1.5 hrs each
4 Support Sessions
4 hrs each

Course Overview

This course bridges the gap between software and hardware. Students learn basic electronics, build circuits on a breadboard, and learn to program an Arduino microcontroller to control real-world outputs. The course culminates in each student building, wiring, and coding their own robot.

By the End, Students Will:

- Understand basic electronics concepts and components
- Build working circuits on a breadboard
- Program an Arduino to read inputs and control outputs
- Assemble, test, and debug a real robot project

#	Session / Topic	Activity / Project	Goal / Principle
1	Intro to Electronics	Voltage, current, resistance — live demo	Understand electricity fundamentals
2	Components + Circuits	Breadboard: LEDs, resistors, series & parallel	Hands-on circuit building
TS 1	**Circuit Building Lab**	Free build on breadboard — try new circuits	*Confidence with physical components*
3	Sensors + Microcontrollers	Buttons, sensors + Arduino introduction	Read digital inputs
4	Programming Outputs	Blink LED, control a motor via code	Connect code to hardware
TS 2	**Sensor & Code Exercises**	Read sensor values, respond with outputs	*Applied coding on hardware*
5	Motor Control	Motor drivers, direction and speed logic	Control real-world actuation
6	Robot Design + Planning	Sketch robot plan, choose components	Engineering and design thinking
TS 3	**Design Review + Component Prep**	Finalise robot plan, test components	*Readiness to build*
7	Build Your Robot	Assemble, wire, and code the robot	Full system integration
8	Test, Debug + Present	Fix issues, improve performance, demo robot	Real product delivery
TS 4	**Robot Demo Day**	Each student presents their robot to the group	*Showcase and celebration*

■ Core Session — 1.5 hrs (Engineer-led) ■ Technical Support Session — 4 hrs (CTA-led)

⚡ Build a Web App with AI

Ages 13+ · 500 EGP · AI Tools (Lovable / v0) · Project: Live Web App

8 Core Sessions
1.5 hrs each
4 Support Sessions
4 hrs each

Course Overview

This course teaches students to think like product builders. Using AI-powered tools like Lovable and v0, students design, build, and deploy a real web application — from wireframe to live URL — in 8 sessions. No prior web knowledge required. The focus is on product thinking, prompt engineering, and shipping.

By the End, Students Will:

- Use AI tools to generate and iterate on web app features
- Design a product from wireframe to working prototype
- Connect data and forms to create functional applications
- Deploy a live web app accessible by real URL

#	Session / Topic	Activity / Project	Goal / Principle
1	What is a Web App?	Explore real apps — pages, features, data flow	Product thinking and context
2	Intro to AI Tools	First prompt in Lovable — generate a landing page	Understand prompt-to-product flow
TS 1	**Explore & Customise**	Tweak generated pages — layout, text, colours	*Confidence with AI tools*
3	Designing Your App Idea	Sketch wireframe + write feature list	Product design and planning
4	Prompting for Features	Add forms, buttons, navigation with AI prompts	Iterative and precise prompting
TS 2	**Feature Building Lab**	Each student builds one new feature independently	*Independence and creativity*
5	Connecting Data	Add form submissions and simple data storage	Understand data flow in apps
6	UI Polish + Branding	Colours, fonts, logo — make the app yours	Visual identity and UX thinking
TS 3	**Design Review + Bug Fixing**	Peer feedback session — improve each other's apps	*Collaboration and refinement*
7	Deploying Your App	Publish live — every student gets a real URL	Ship it — real-world launch
8	Final Project Demo	Present live app to the group + get feedback	Full product launch mindset
TS 4	**App Showcase Day**	Live demos, iterate on feedback, celebrate	*Pride and launch culture*

■ Core Session — 1.5 hrs (Engineer-led) ■ Technical Support Session — 4 hrs (CTA-led)

ElSewedy GenZ Coders · Phase 1 Curriculum · Confidential Internal Document · 2026